

YILAN CITY, Taiwan

WMO: 467080

Lat: 24.7639N

Lon: 121.7565E

Elev: 24

StdP: 14.68

Time Zone: 8.00 (E08)

Period: 90-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	50.9	52.4	43.4	41.6	53.0	46.4	46.7	54.6	17.5	62.1	15.6	62.5	4.1	250	0.264

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	92.2	82.5	91.2	82.2	89.9	81.6	84.4	90.2	83.4	89.3	82.4	88.3	7.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.7	170.5	88.9	82.0	166.5	88.4	80.7	159.7	87.2	48.7	90.7	47.4	89.3	46.5	88.6	91.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.7	13.9	11.4		DB	47.6	96.6	2.2	2.9	46.0	98.7	44.7	100.4	43.5	102.0	41.9	104.1
				WB	44.2	87.6	2.9	2.0	42.1	89.0	40.3	90.2	38.7	91.3	36.6	92.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	73.1	61.5	62.2	66.5	71.6	76.1	80.8	84.1	83.7	80.3	74.5	70.0	64.9
	DBStd	8.90	4.72	5.52	5.51	5.69	4.15	3.58	1.79	1.93	3.25	3.95	4.62	4.78
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	375	127	112	51	14	1	0	0	0	0	0	10	60
	CDD50	8426	357	341	511	648	810	925	1056	1045	909	760	601	463
	CDD65	3327	19	33	97	212	346	475	591	580	459	295	161	59
	CDH74	28353	22	65	288	1041	2204	4644	7064	6745	4245	1442	502	91
CDH80	9482	1	2	23	184	504	1554	2948	2665	1277	253	66	5	
Wind	WSAvg	3.5	3.5	3.3	3.3	3.0	2.9	2.9	4.4	4.3	4.3	3.7	3.6	3.1
Precipitation	PrecAvg	128.9	9.3	9.3	6.9	7.0	9.3	9.5	6.0	10.0	17.4	18.4	15.6	10.3
	PrecMax	211.4	24.1	23.4	10.2	18.0	22.7	19.6	15.9	20.4	53.7	67.1	40.2	23.8
	PrecMin	70.6	2.6	1.6	2.1	1.5	2.2	2.6	0.2	2.7	2.7	2.4	2.3	2.1
	PrecStd	30.7	5.0	5.1	1.9	4.2	5.3	3.6	4.2	5.8	10.9	13.7	10.1	6.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.8	79.1	82.5	86.4	89.2	91.8	93.6	93.6	91.8	87.6	84.6	80.4
		MCWB	69.2	72.3	74.6	79.1	81.7	82.4	82.4	83.1	82.2	78.6	77.6	74.4
	2%	DB	73.5	76.0	79.8	84.2	87.0	90.2	92.4	91.9	89.8	84.7	82.0	76.6
		MCWB	67.5	70.2	73.2	77.3	80.1	81.8	82.4	82.7	81.0	77.6	74.9	70.8
	5%	DB	71.2	73.3	77.2	82.3	84.6	89.3	91.4	91.1	88.1	82.9	79.1	73.7
		MCWB	65.9	67.7	71.2	76.3	78.6	81.5	82.1	82.4	80.5	76.1	72.9	69.1
	10%	DB	68.2	70.1	74.9	79.9	82.7	87.6	89.9	89.6	86.4	80.7	76.7	71.5
		MCWB	64.3	65.9	69.9	74.7	77.4	80.8	81.6	81.7	79.8	74.9	71.5	67.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.5	73.5	75.8	80.1	82.5	84.7	85.9	85.3	84.4	80.9	78.6	75.0
		MCDB	74.8	78.4	81.0	84.9	87.5	89.9	90.9	90.8	89.8	85.2	83.6	78.8
	2%	WB	68.6	70.6	73.9	78.3	80.6	82.8	84.4	84.3	82.6	79.0	76.0	71.8
		MCDB	72.7	74.8	78.8	83.1	85.6	88.5	90.3	90.4	87.6	83.1	80.3	74.7
	5%	WB	66.5	68.4	72.1	76.6	79.2	81.9	83.1	83.3	81.5	77.6	74.1	69.9
		MCDB	70.0	72.3	76.3	81.5	84.0	87.6	89.4	89.3	86.6	81.0	77.8	72.6
	10%	WB	64.8	66.2	70.3	75.0	78.0	80.8	82.2	82.3	80.4	76.3	72.6	68.3
		MCDB	67.8	69.3	73.9	79.5	82.4	86.4	88.6	88.1	85.5	79.3	75.9	71.4
Mean Daily Temperature Range	5% DB	MDBR	10.1	9.9	11.2	11.5	11.6	11.8	13.3	12.5	11.6	9.8	10.8	9.9
		MCDBR	14.9	15.7	15.5	14.4	14.0	13.8	14.5	14.1	13.8	13.8	14.4	14.5
		MCWBR	11.1	11.7	11.1	10.3	9.8	8.3	8.9	8.6	9.2	9.5	10.3	10.9
	5% WB	MCDBR	13.6	14.7	15.1	13.7	13.2	13.2	14.0	13.2	12.9	11.4	13.3	13.3
MCWBR		10.7	11.8	11.4	10.6	9.7	8.6	9.7	9.0	9.4	9.1	10.4	10.8	
Clear-Sky Solar Irradiance	taub		0.541	0.572	0.668	0.673	0.593	0.553	0.518	0.534	0.521	0.528	0.529	0.542
	taud		1.898	1.864	1.674	1.715	1.935	2.050	2.156	2.101	2.120	2.039	2.008	1.918
	Ebn at Noon		222	227	213	216	233	241	250	246	246	236	225	215
	Edh at Noon		56	62	78	76	61	54	49	51	49	51	49	53
All-Sky Solar Radiation	RadAvg		710	817	992	1120	1314	1538	1860	1682	1450	1156	831	702
	RadStd		128	152	147	171	224	181	109	150	151	139	99	117

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)	N/A	N/A	N/A	+0.58	N/A	N/A	N/A	N/A	N/A	+146 (47)

Nomenclature: See separate page